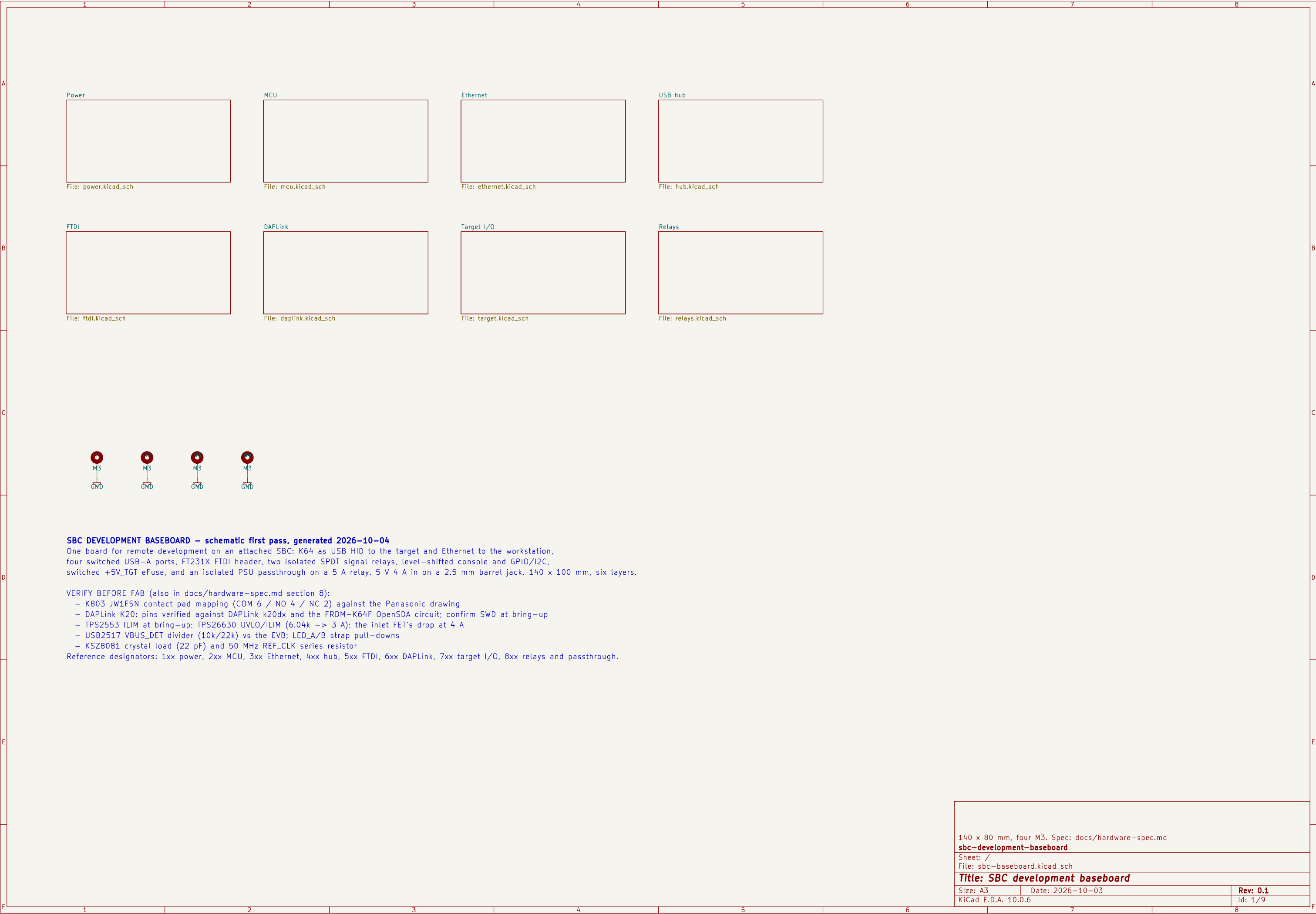




M3

GND

M3

GND

M3

GND

M3

GND

SBC DEVELOPMENT BASEBOARD – schematic first pass, generated 2026–10–04

One board for remote development on an attached SBC: K64 as USB HID to the target and Ethernet to the workstation, four switched USB–A ports, FT231X FTDI header, two isolated SPDT signal relays, level–shifted console and GPIO/I2C, switched +5V_TGT eFuse, and an isolated PSU passthrough on a 5 A relay. 5 V 4 A in on a 2.5 mm barrel jack. 140 x 100 mm, six layers.

VERIFY BEFORE FAB (also in docs/hardware–spec.md section 8):

- K803 JW1FSN contact pad mapping (COM 6 / NO 4 / NC 2) against the Panasonic drawing
- DAPLink K20: pins verified against DAPLink k20dx and the FRDM–K64F OpenSDA circuit; confirm SWD at bring–up
- TPS2553 ILIM at bring–up; TPS26630 UVLO/ILIM (6.04k –> 3 A); the inlet FET’s drop at 4 A
- USB2517 VBUS_DET divider (10k/22k) vs the EVB; LED_A/B strap pull–downs
- KSZ8081 crystal load (22 pF) and 50 MHz REF_CLK series resistor

Reference designators: 1xx power, 2xx MCU, 3xx Ethernet, 4xx hub, 5xx FTDI, 6xx DAPLink, 7xx target I/O, 8xx relays and passthrough.

140 x 80 mm, four M3. Spec: docs/hardware–spec.md

sbc–development–baseboard

Sheet: /

File: sbc–baseboard.kicad_sch

Title: SBC development baseboard

Size: A3

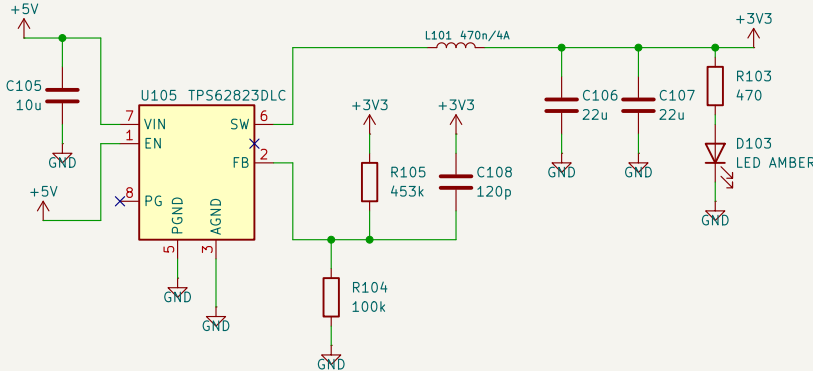
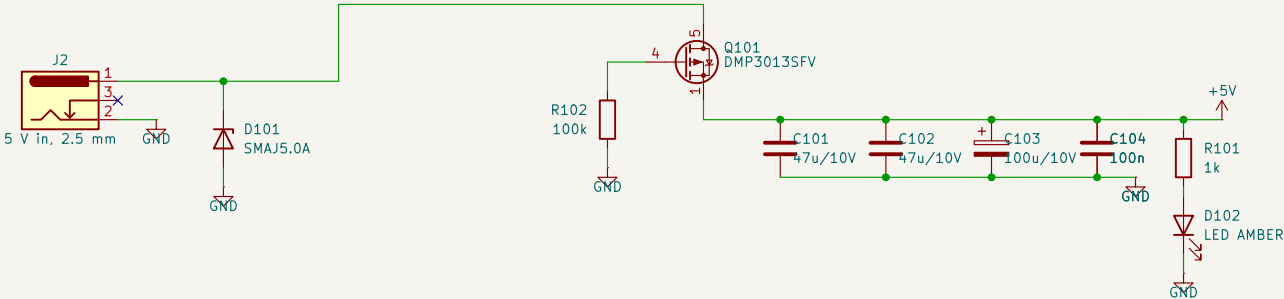
Date: 2026–10–03

Rev: 0.1

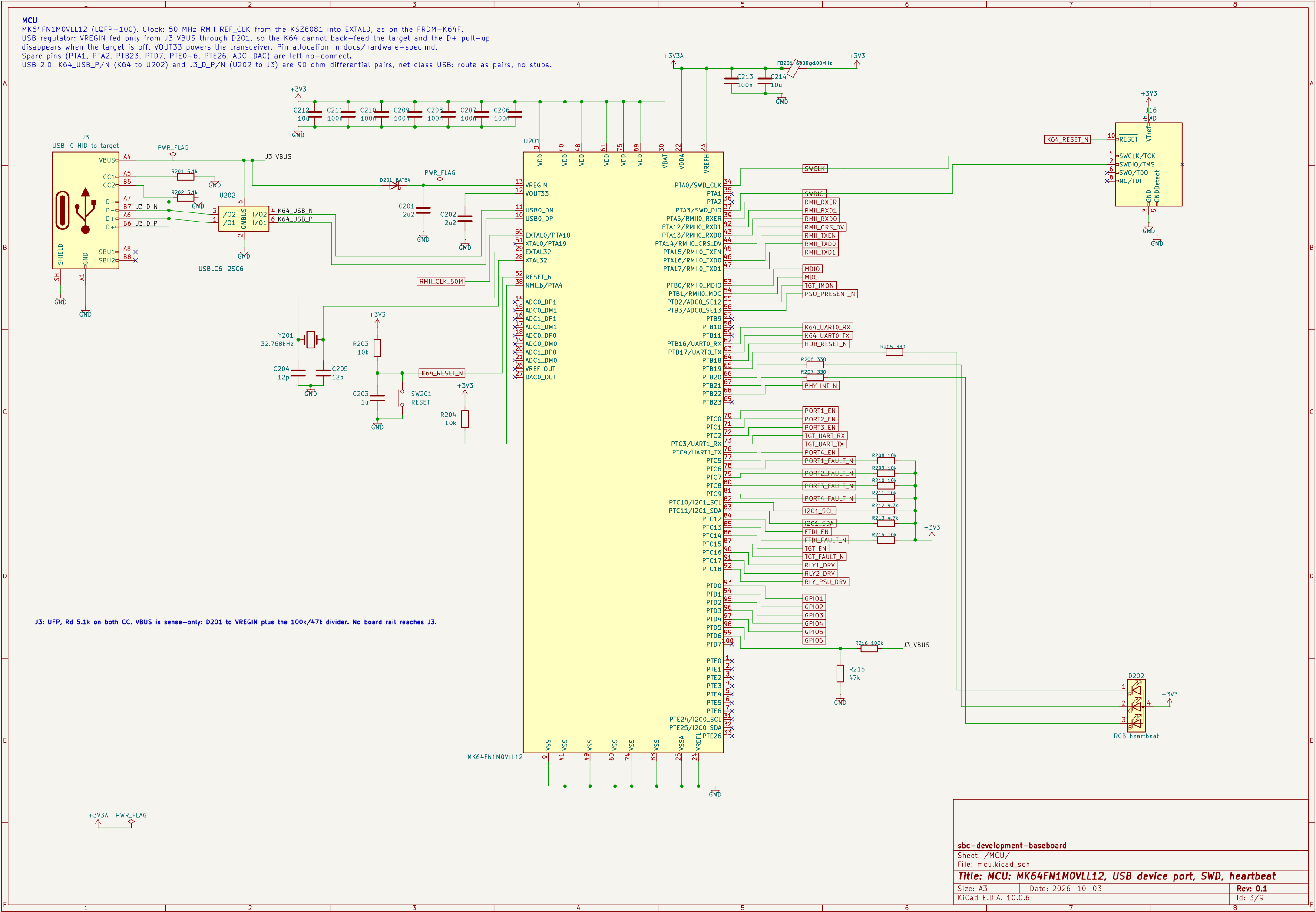
KiCad E.D.A. 10.0.6

Id: 1/9

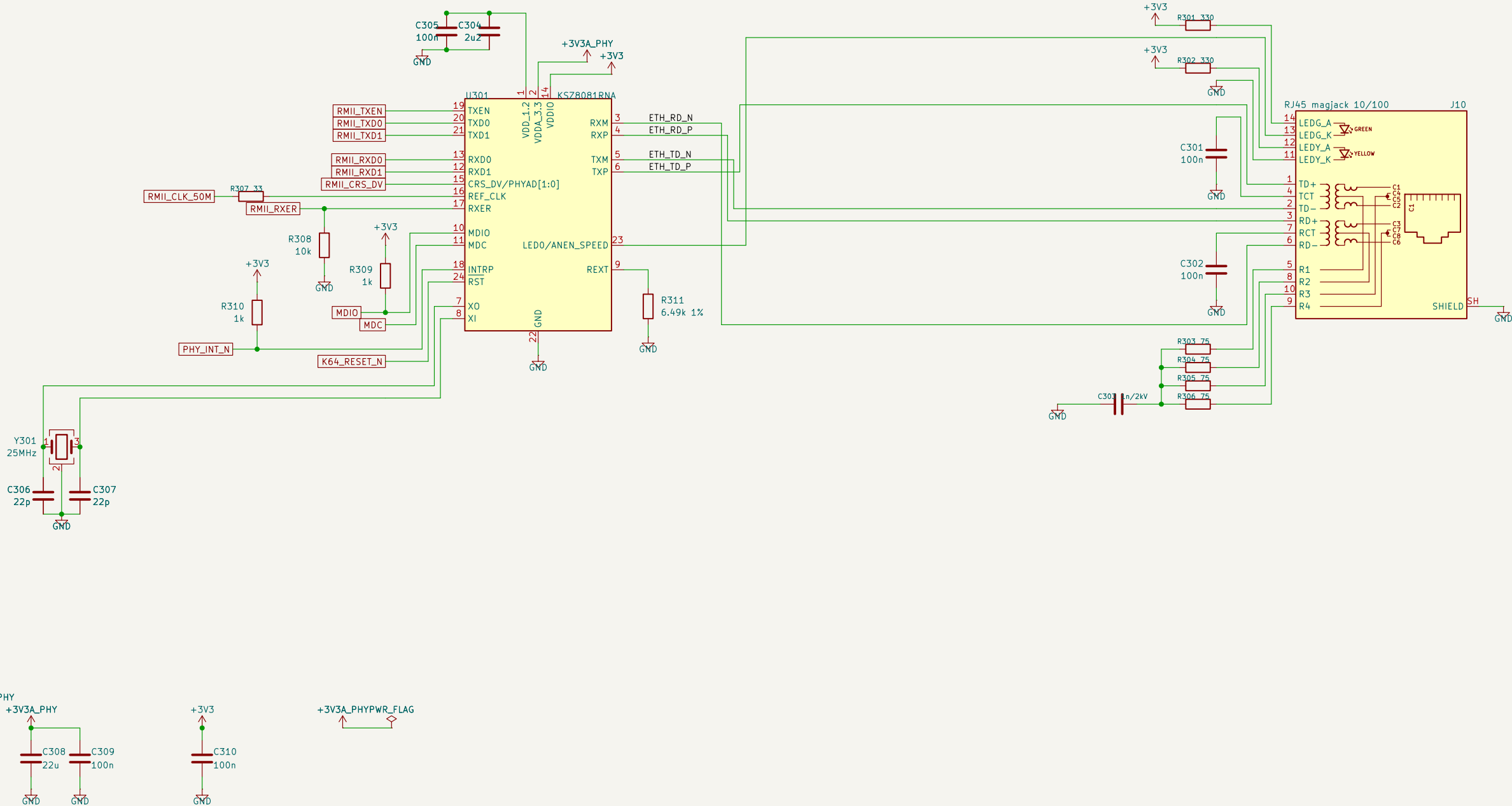
POWER
J2: 2.5 x 5.5 mm barrel jack, centre positive, for a 5 V 4 A (20 W) adapter (2026-10-10; a USB-C PD inlet with two 5 V bucks until then). SMAJ5.0A clamps the inlet; Q101, a P-channel MOSFET with its body diode toward the rail and its gate at GND, blocks reverse polarity at 9.5 mOhm, 2 x 47 uF + 100 uF bulk behind it; the hub's downstream ports want 120 uF. No series fuse: the adapter limits the inlet, the eFuse the target rail, the TPS2553s the ports. The adapter's 4 A is the only limit on the sum: the ports (4 x 1.1 A), the target rail (3 A) and the board (0.5 A) can ask for more together; firmware keeps the budget (docs/hardware-spec.md section 4). +5V is laid out for 4 A (class PWR_4A, 2.5 mm or pours); +5V_TGT leaves the eFuse at up to 3 A in the same class.



MCU
MK64FN1M0VLL12 (LQFP-100). Clock: 50 MHz RMII REF_CLK from the KSZ8081 into EXTAL0, as on the FRDM-K64F.
USB regulator: VREGIN fed only from J3 VBUS through D201, so the K64 cannot back-feed the target and the D+ pull-up disappears when the target is off. VOUT33 powers the transceiver. Pin allocation in docs/hardware-spec.md.
Spare pins (PTA1, PTA2, PTB23, PTD7, PTE0-6, PTE26, ADC, DAC) are left no-connect.
USB 2.0: K64_USB_P/N (K64 to U202) and J3_D_P/N (U202 to J3) are 90 ohm differential pairs, net class USB: route as pairs, no stubs.



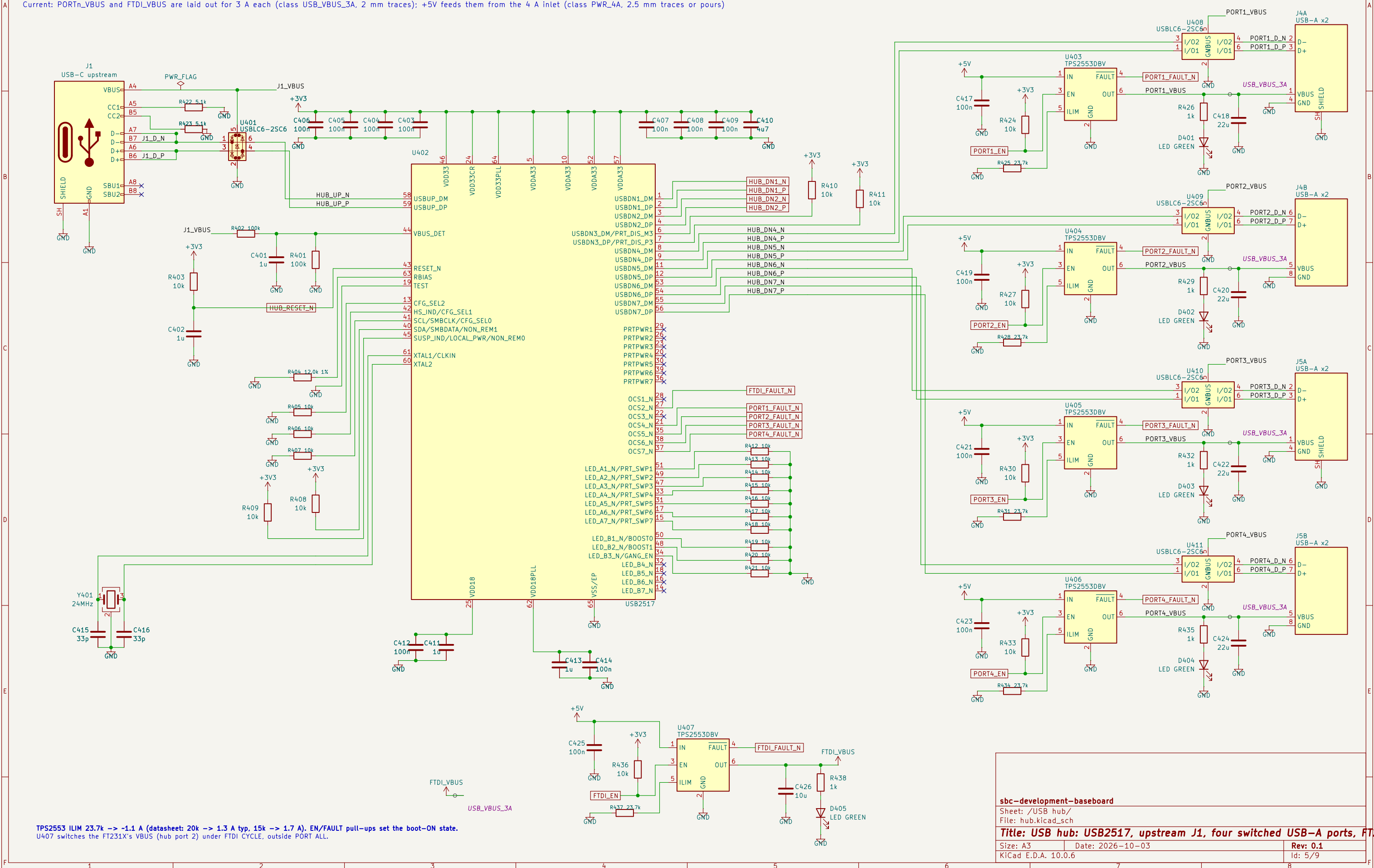
ETHERNET
KSZ8081RNA in RMII 25 MHz mode (its power-up default): 25 MHz crystal on XI/XO, REF_CLK outputs 50 MHz to the K64 EXTAL0 through R301. This is the FRDM-K64F arrangement (Zephyr: microchip,interface-type = rmii-25MHz). PHY address 0 (pins 15/16 internal pull-downs; the MAC does not drive them). RXER needs the external pull-down. Transformer centre taps go to 0.1 uF each, not to a supply (datasheet 11.x). Bob Smith: 4x75R to 1 nF/2 kV.



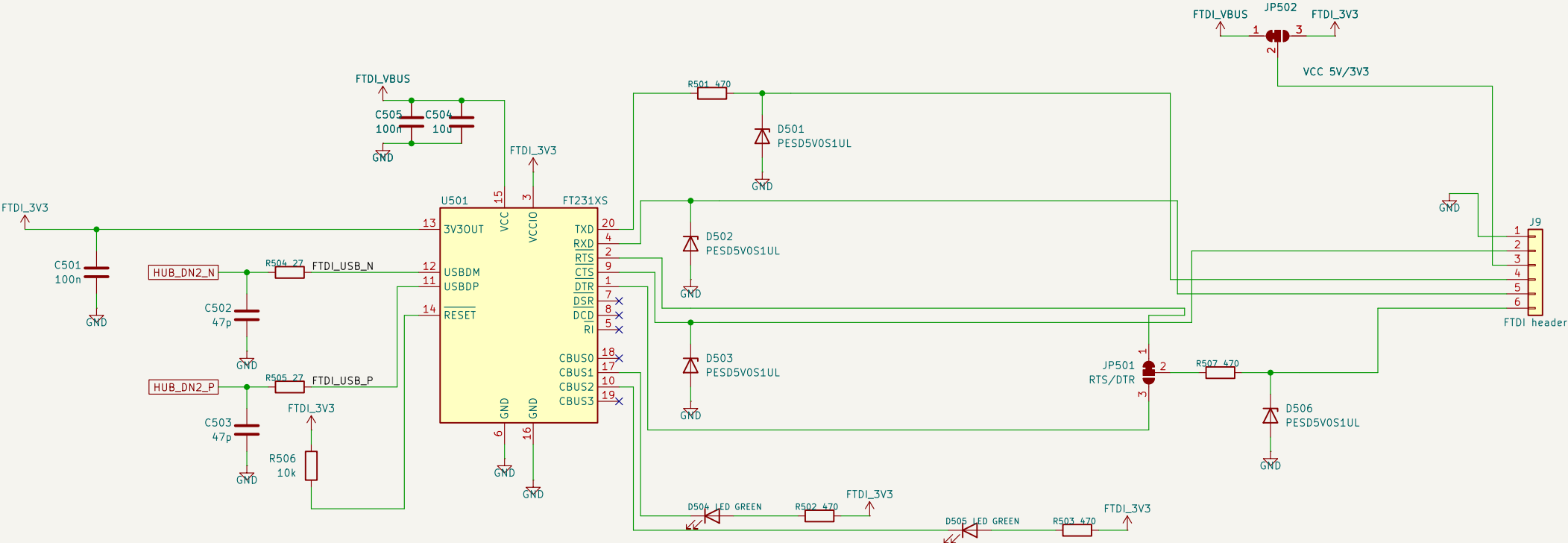
Crystal: 25 MHz +/-50 ppm, CL 16 pF typ (datasheet) -> 22 pF each, trim at bring-up.
Green LED = link/activity (PHY LED0, active low). Yellow LED = PHY powered (cathode to GND).

USB HUB

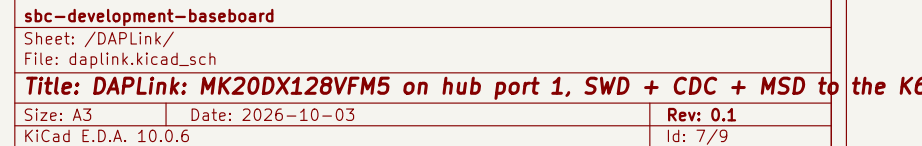
USB2517 strap-configured (CFG_SEL=000), no firmware needed for the USB tree to come up.
Port map: 1 = DAPLink, 2 = FT231X, 3 = disabled (DN3 pulled up), 4..7 = USB-A PORT1..PORT4 (J4A/J4B = PORT1/PORT2 on hub ports 4,5; J5A/J5B = PORT3/PORT4 on ports 6,7).
NON_REM=11 (pin 40 up, pin 45 up) marks ports 1-3 non-removable and LOCAL_PWR high reports self-powered.
LED_A/B pins pulled down: PRT_SWP normal polarity, BOOST=00, GANG_EN=0 (individual over-current).
Port power is switched by TPS2553 under K64 GPIO (boot ON via pull-ups); PRT_PWR outputs unused; /FAULT feeds OCSx_N.
USB 2.0 pairs are named *_P / *_N (KiCad's differential-pair convention) and sit in net class USB: 90 ohm differential, routed as pairs, no stubs. Each ESD array splits its pair into a hub side (HUB_DNn) and a connector side (PORTn_D). Set the class's width and gap from the stackup before routing.
Current: PORTn_VBUS and FTDI_VBUS are laid out for 3 A each (class USB_VBUS_3A, 2 mm traces); +5V feeds them from the 4 A inlet (class PWR_4A, 2.5 mm traces or pours)



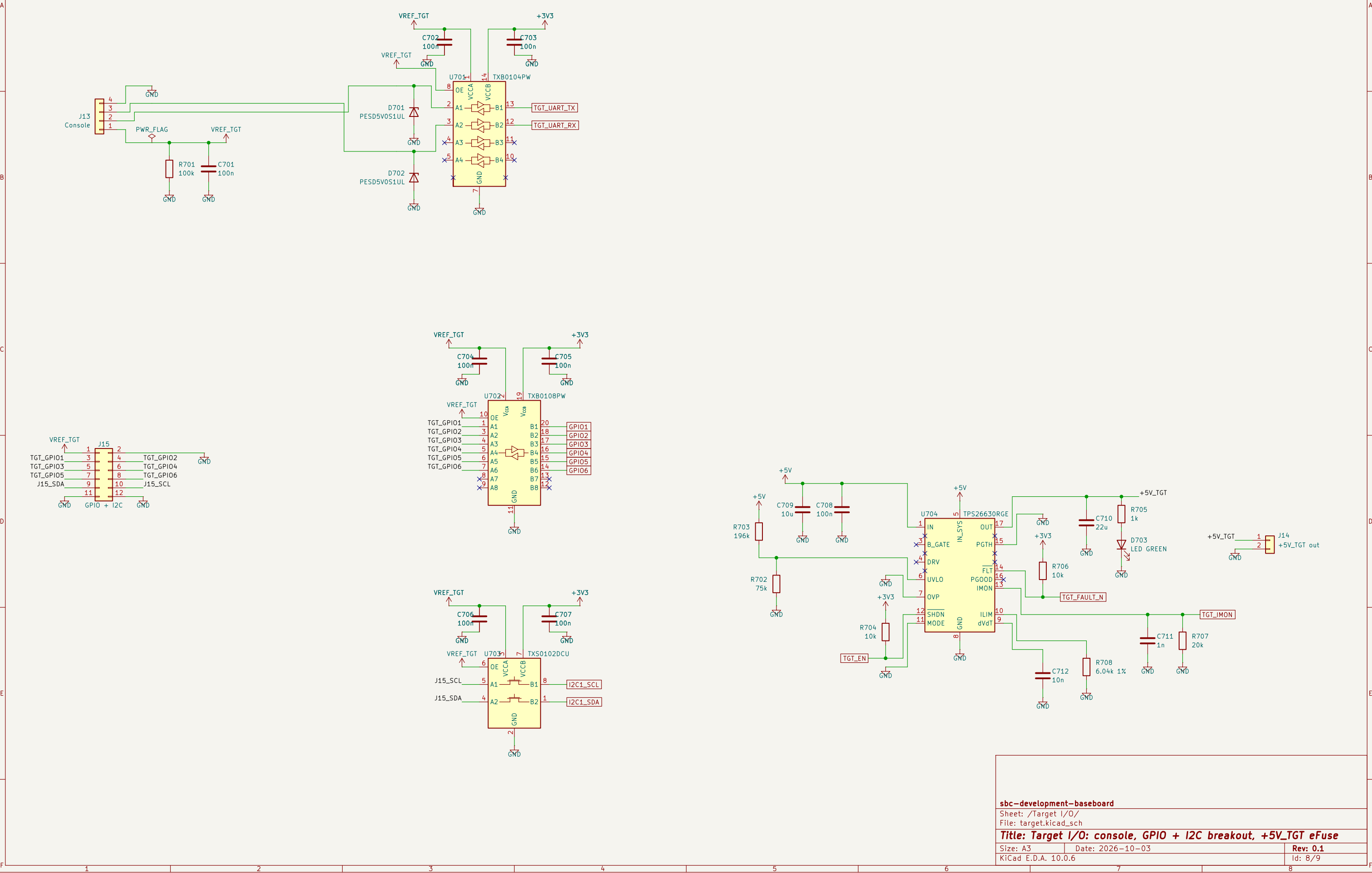
FTDI HEADER J9
Pinout = Adafruit FTDI Friend (the TTL-232R order): 1 GND (blk) 2 CTS (brn) 3 VCC (red) 4 TXD (org) 5 RXD (yel) 6 RTS/DTR (grn).
TXD/RXD named from the FT231X side: TXD drives the target's RX. 3.3 V levels only; 1.8 V consoles use J13.
JP501: pin 6 = RTS# (A-C, bridged by default) or DTR# (cut, bridge B-C). JP502: pin 3 VCC = FTDI_VBUS 5 V (A-C, default) or 3V3OUT 3.3 V / 50 mA (B-C).
VCC comes from the TPS2553 on the hub sheet (FTDL_VBUS): FTDI CYCLE power-cycles the bridge without a replug.
CBUS defaults: CBUS1 = RXLED#, CBUS2 = TXLED# (factory MTP); change with FT_PROG if ever needed.



MK20DX128VFM5 running DAPLink (k20dx.HC), powered from +5V through its own USB regulator (VOUT33 -> VDD).
 Pin use includes the DAPLink k20dx.HC: PTC5 = SWCLK, PTC6 = SWDIO, PTB1 = nRESET to the K64, PTD4 = LED.
 UART1 PTC3/PTC4 bridges to the K64 UART0 (CDC console). Pins verified against DAPLink k20dx_IQ_Config.h and the FRDM-K64F OpenSDA circuit:
 PTC7 (SWDIO_IN) ties to SWDIO, 33 R in series with USB D+/D- (FRDM R20/R22). No level shifters: the K64 target is 3.3 V like the K20.
 J601 provides the K20 itself (bootloader + DAPLink). SW601 held at power-up enters DAPLink maintenance mode.

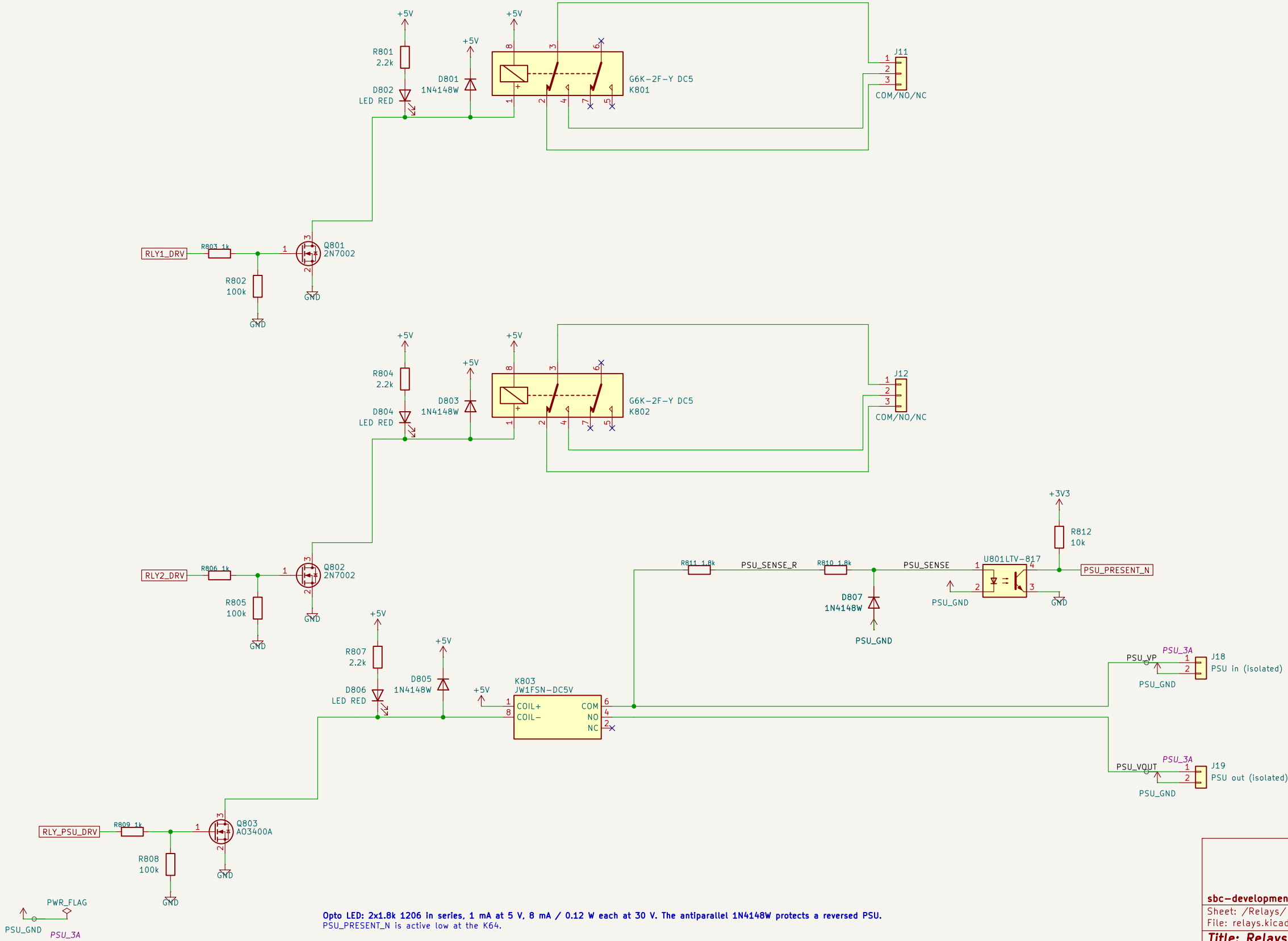


TARGET-FACING I/O
VREF_TGT comes from the target on J13.1 / J15.1 and sets the level of every translator (TXB0104/TXB0108/TXS0102).
OE pins follow VREF_TGT: with the target off the translators are high-impedance. J15 GPIO are push-pull (TXB).
TXS0102 instead of PCA9306: works with VREF_TGT equal to 3.3 V (PCA9306 needs VREF2 > VREF1). Range 1.65–3.6 V.
TPS26630: ILIM 6.04k -> 3 A (RILIM = 18 kOhm.A / IOL; 3.65k -> 5 A until 2026-10-10, when the 20 W inlet set the budget). UVLO 196k/75k -> 4.3 V, OVP/MODE/PGTH to GND (latch-off). IMON 20k -> 1.7 V at 3 A into PTB2.



RELAYS AND PASSTHROUGH

K801/K802 G6K-2F-Y: pole 1 used (COM 3, NC 2, NO 4), pole 2 unused. Coils boot de-energized (gate pull-downs).
Red LEDs sit across each coil and show the energized state. Flyback 1N4148W on every coil.
Passthrough J18 -> K803 (NO contact) -> J19 on nets PSU_VP / PSU_VOUT / PSU_GND that touch no board net: PSU_GND is its own ground.
Only the optocoupler's emitter sits across the passthrough, sized for 5-30 V. No indicator LED there.
K803 JW1FSN pad map COM=6 NO=4 NC=2: VERIFY against the Panasonic terminal drawing before fab.
Current: the passthrough J18 -> K803 -> J19 (PSU_VP, PSU_VOUT, PSU_GND) is laid out for 3 A continuous: class PSU_3A, 2 mm traces or pours, two vias per layer change.



Opto LED: 2x1.8k 1206 in series, 1 mA at 5 V, 8 mA / 0.12 W each at 30 V. The antiparallel 1N4148W protects a reversed PSU.
PSU_PRESENT_N is active low at the K64.

sbc-development-baseboard

Sheet: /Relays/
File: relays.kicad_sch

Title: Relays: two SPDT signal relays, isolated PSU passthrough

Size: A3	Date: 2026-10-03	Rev: 0.1
KiCad E.D.A. 10.0.6		Id: 9/9